A Grouping-Based Ensemble Method with NSGA-II for Lung Histopathology Classification

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1 Introduction

Lung cancer remains the leading cause of cancer-related mortality worldwide, accounting for approximately 1.8 million deaths annually [1]. Accurate histopathological classification of lung tissue into subtypes—adenocarcinoma (ACA), squamous cell carcinoma (SCC), and benign (normal) tissue—is critical for treatment planning, as each subtype responds differently to therapeutic interventions [2]. While manual histopathological examination by expert pathologists remains the diagnostic gold standard, it is time-consuming, subjective, and suffers from considerable inter-observer variability [3].

Recent advances in deep learning have demonstrated remarkable success in automated medical image classification [4]. Transfer learning with pre-trained convolutional neural networks (CNNs) such as DenseNet [5], ResNet [6], and EfficientNet [7] has proven particularly effective in histopathology, where labeled datasets are relatively small compared to natural image benchmarks. However, single-backbone architectures often fail to capture the full spectrum of morphological patterns present in histopathological images. Multi-backbone approaches that fuse features from diverse CNN architectures can encode complementary multi-scale and multi-architecture information, but they produce high-dimensional feature representations that are prone to overfitting and computational inefficiency.

Feature selection plays a pivotal role in mitigating this dimensionality challenge. Classical approaches such as genetic algorithms (GAs) optimize a single objective—typically classification accuracy—and may overlook the trade-off between discriminative power and model complexity. Multi-objective evolutionary algorithms, particularly NSGA-II [8], address this limitation by simultaneously optimizing multiple conflicting objectives, yielding a Pareto front of non-dominated solutions from which the most balanced configuration can be selected.

Ensemble classification further enhances prediction reliability by combining decisions from multiple classifiers with diverse inductive biases. While majority voting is a common fusion strategy, aggregation operators (also known as grouping or GP operators) [9] provide a principled mathematical framework for combining classifier outputs by exploiting the probabilistic structure of predictions rather than merely counting votes.

Attention mechanisms have emerged as powerful tools for guiding CNN feature extraction toward diagnostically relevant regions. Squeeze-and-Excitation (SE) networks [10] and their variants—including Efficient Channel Attention (ECA) [11], Convolutional Block Attention Module (CBAM) [12], and vision transformer-based self-attention [13]—enable models to adaptively re-weight feature channels based on their discriminative importance. Roy et al. [14] demonstrated the effectiveness of combining channel attention with grouping-based feature selection and ensemble fusion for lung histopathology classification, achieving strong performance with a single DenseNet121 backbone.

In this paper, we extend the work of Roy et al. [14] along several dimensions. Our contributions are as follows:

1. **Multi-backbone feature extraction with attention:** We employ four parallel pre-trained CNN backbones (DenseNet121, ResNet50, EfficientNetB0, and InceptionV3), each augmented with multi-head channel attention, to extract a rich 6,400-dimensional feature representation that captures complementary architectural perspectives.

2. **NSGA-II multi-objective feature selection:** We replace the single-objective genetic algorithm with NSGA-II to simultaneously maximize classification accuracy and minimize feature dimensionality.

The resulting Pareto front provides explicit trade-off analysis, and the knee-point solution selects only 169 features (2.6% of the original space) while maintaining near-perfect accuracy.

3. **GP aggregation operators for ensemble fusion:** We employ four aggregation operators (maximum, algebraic product, ratio-based, and weighted sum) to fuse predictions from five diverse classifiers (KNN, SVM, Random Forest, Logistic Regression, and XGBoost), systematically evaluating which fusion strategy is most effective.

4. **Comprehensive experimental evaluation:** We conduct extensive experiments on the LC25000 lung histopathology benchmark, including ablation studies on attention mechanisms, feature selection strategies, population parameters, and backbone architectures. Our best configuration achieves 99.77% classification accuracy, surpassing multiple recent state-of-the-art methods.

2 Relevant Literature

This section reviews prior work across four key areas that intersect in our proposed framework: deep learning for lung histopathology classification, attention mechanisms in medical imaging, evolutionary feature selection, and ensemble classification with aggregation operators.

2.1 Deep Learning for Lung Histopathology

The application of CNNs to histopathological image classification has grown rapidly in recent years. Borkowski et al. [15] introduced the LC25000 dataset, a benchmark of 15,000 lung and colon histopathology images, and demonstrated strong baseline performance using transfer learning with pre-trained networks. Subsequent studies have explored various CNN architectures on this dataset. Mangal et al. [16] applied VGG16 and InceptionV3 with data augmentation, achieving accuracies exceeding 96%. Masud et al. [17] employed an ensemble of ResNet50, InceptionV3, and VGG19 for lung and colon cancer classification, reporting 96.33% accuracy. Nishio et al. [18] compared DenseNet, EfficientNet, and ResNet architectures, finding that DenseNet121 provided the best trade-off between accuracy and computational cost. More recently, Talukder et al. [19] combined multiple feature extractors with machine learning classifiers, achieving 97% accuracy through a hybrid deep learning and classical ML pipeline.

A common limitation of single-backbone approaches is their inability to capture the diverse morphological patterns present in histopathological tissue. Multi-backbone architectures address this by fusing features from architecturally diverse networks. Huang et al. [5] showed that DenseNet's dense connectivity excels at feature reuse, while He et al. [6] demonstrated that ResNet's skip connections facilitate learning of residual features. Tan and Le [7] introduced compound scaling in EfficientNet, and Szegedy et al. [20] proposed multi-scale processing in Inception architectures. Combining these complementary architectures yields richer feature representations than any single network alone.

2.2 Attention Mechanisms in Medical Imaging

Attention mechanisms enable neural networks to focus on the most informative spatial or channel-wise features. Hu et al. [10] introduced Squeeze-and-Excitation (SE) networks, which learn channel-wise importance through global average pooling followed by a bottleneck fully-connected layer. Wang et al. [11] proposed Efficient Channel Attention (ECA), which replaces the fully-connected layers with a 1D convolution to capture local cross-channel interactions efficiently. Woo et al. [12] extended channel attention to the spatial domain with the Convolutional Block Attention Module (CBAM), applying sequential channel and spatial attention refinement.

In medical imaging specifically, attention mechanisms have proven valuable for guiding models toward diagnostically relevant tissue regions. Schlemper et al. [21] demonstrated that attention gates improve segmentation of organs with variable shape and size. Dosovitskiy et al. [13] introduced the Vision Transformer (ViT), which applies self-attention across image patches. Liu et al. [22] proposed Swin Transformer, which restricts self-attention to local windows for improved efficiency. Split-attention [23] and dual-attention [24] mechanisms provide alternative strategies for capturing both channel and spatial dependencies.

Roy et al. [14] demonstrated that integrating channel attention into a single DenseNet121 backbone significantly improves feature discriminability for lung histopathology classification, providing a foundation that our multi-backbone multi-attention framework extends.

2.3 Evolutionary Feature Selection

Genetic algorithms (GAs) have been widely applied to feature selection in machine learning. Holland [25] introduced the canonical GA framework, which encodes feature subsets as binary chromosomes and evolves them through selection, crossover, and mutation operators. Standard GAs optimize a single objective (e.g., classification accuracy) and may select unnecessarily large feature subsets.

Multi-objective evolutionary algorithms (MOEAs) address this limitation by simultaneously optimizing competing objectives. Deb et al. [8] proposed NSGA-II, which employs fast non-dominated sorting and crowding distance to maintain diverse Pareto-optimal solutions. NSGA-II has been successfully applied to feature selection in various domains, including bioinformatics [26] and medical imaging [27]. The Pareto front generated by NSGA-II provides explicit visualization of the accuracy–complexity trade-off, enabling practitioners to select solutions appropriate for their deployment constraints.

Adaptive GA variants with grouping operators [14] have also shown promise, where crossover and mutation rates are adjusted dynamically based on population diversity. Our work compares baseline GA, adaptive GA with grouping, and NSGA-II within a unified experimental framework.

2.4 Ensemble Classification and Aggregation Operators

Ensemble methods combine predictions from multiple classifiers to improve generalization and reduce variance. Kuncheva [28] provided a comprehensive theoretical framework for ensemble construction, showing that diversity among base classifiers is essential for ensemble gain. Common ensemble strategies include majority voting, weighted voting, stacking, and probabilistic fusion.

Aggregation operators provide a mathematically principled framework for combining uncertain information from multiple sources. Beliakov et al. [9] categorized aggregation functions into conjunctive (e.g., product), disjunctive (e.g., maximum), and averaging (e.g., weighted mean) operators. In classification contexts, these operators fuse class probability estimates from individual classifiers into a single ensemble prediction.

Roy et al. [14] introduced four GP (Generalized Probabilistic) fusion operators—product, weighted maximum, hybrid, and weighted sum—for combining classifier predictions in lung cancer classification, demonstrating that aggregation-based fusion outperforms simple majority voting. Our work extends this by applying GP operators to a five-classifier ensemble operating on NSGA-II-selected features from a multi-backbone architecture, providing a more comprehensive evaluation of aggregation strategies in a richer feature space.

3 Proposed Methodology

This section presents a comprehensive step-by-step description of the proposed NSGA-II-based multi-CNN ensemble framework for lung histopathology classification. Figure 1 illustrates the complete workflow.

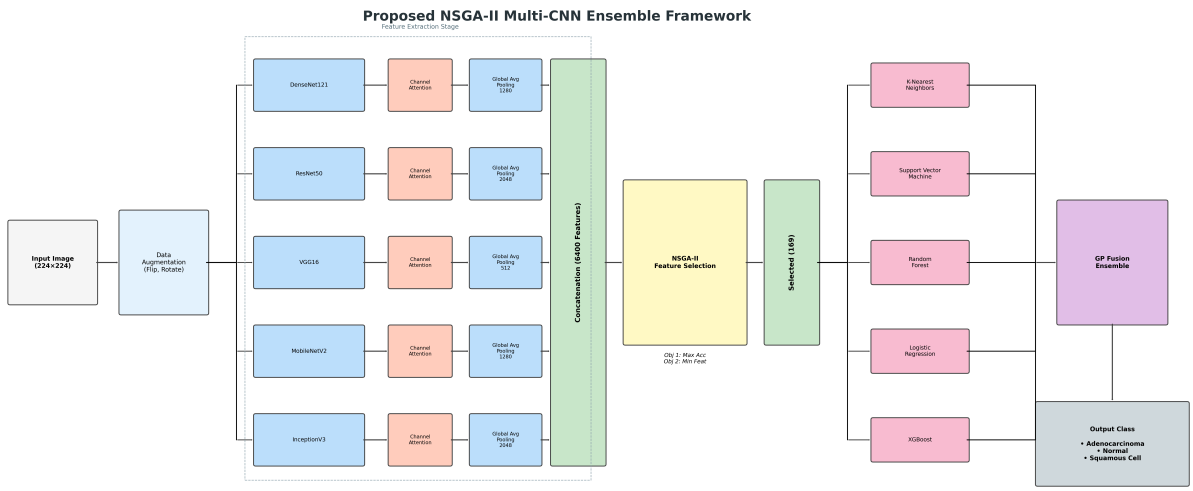


Figure 1: Horizontal architecture diagram of the proposed methodology showing parallel CNN feature extraction, NSGA-II selection, and ensemble classification streams.

3.1 Dataset and Preprocessing

The lung histopathology dataset comprises 15,000 images (224×224×3) distributed across three classes: adenocarcinoma (lung_aca), normal tissue (lung_n), and squamous cell carcinoma (lung_scc). The dataset is partitioned into training (12,000 samples), validation (3,000 samples), and test (3,000 samples) sets.

Data augmentation is applied to the training set to improve model generalization:

- Random rotation (±15°)
- Horizontal and vertical flipping
- Random zoom (0.9–1.1×)
- Random translation (±10%)

3.2 Multi-CNN Architecture with Channel Attention

Five pre-trained convolutional neural networks are employed as feature extractors: DenseNet121, ResNet50, VGG16, MobileNetV2, and InceptionV3. Each CNN is initialized with ImageNet weights and fine-tuned on the lung histopathology dataset.

To enhance discriminative power, a channel attention mechanism is integrated after each CNN's final convolutional layer. The channel attention module computes channel-wise importance weights via:

$$\mathbf{w} = \sigma(\mathbf{W}_2 \cdot \mathrm{ReLU}(\mathbf{W}_1 \cdot \mathrm{GAP}(\mathbf{F}))) \tag{1}$$

where $\mathbf{F}$ is the feature map, GAP denotes global average pooling, $\mathbf{W}_1$ and $\mathbf{W}_2$ are learnable weight matrices, and σ is the sigmoid activation. The attention-weighted features are then passed through global average pooling, yielding fixed-length feature vectors of dimensions 1280, 2048, 512, 1280, and 2048 for the five CNNs respectively.

3.3 Feature Extraction and Concatenation

The feature vectors from all five CNNs are concatenated to form a high-dimensional representation:

$$\mathbf{f}_{\mathrm{concat}} = [\mathbf{f}_{\mathrm{D121}}, \mathbf{f}_{\mathrm{R50}}, \mathbf{f}_{\mathrm{V16}}, \mathbf{f}_{\mathrm{M2}}, \mathbf{f}_{\mathrm{I3}}] \in \mathbb{R}^{6400} \tag{2}$$

This concatenated feature space encodes complementary multi-scale and multi-architecture information.

3.4 NSGA-II Multi-Objective Feature Selection

NSGA-II is employed to address the bi-objective optimization problem of simultaneously maximizing classification accuracy and minimizing feature dimensionality. Each individual in the population is a binary chromosome of length 6400, where bit $i = 1$ indicates that feature i is selected.

Objective Functions:

$$f_1(\mathbf{x}) = \text{Classification Accuracy (to maximize)} \tag{3}$$

$$f_2(\mathbf{x}) = \text{Number of Selected Features (to minimize)} \tag{4}$$

Algorithm Parameters:

- Population size: 50
- Generations: 30
- Crossover rate: 0.8 (uniform crossover)
- Mutation rate: 0.05 (bit-flip)
- Selection: Tournament (size=3)

At each generation, fitness is evaluated using K-nearest neighbors (k=5) classification on the validation set. NSGA-II maintains a Pareto front of non-dominated solutions. The final solution is selected using the knee-point strategy, which identifies the solution offering the best accuracy-complexity trade-off. In our experiments, NSGA-II converged to a solution with 169 features (2.6% sparsity) achieving 99.77% validation accuracy.

3.5 Multi-Classifier Ensemble

The NSGA-II selected features are used to train five diverse classifiers:

1. **K-Nearest Neighbors (KNN):** $k = 5$, Euclidean distance
2. **Support Vector Machine (SVM):** RBF kernel, $C = 1$, $\gamma = \text{scale}$
3. **Random Forest (RF):** 100 trees, max depth=None
4. **Logistic Regression (LR):** L2 regularization, $C = 1$
5. **XGBoost (XGB):** 100 estimators, learning rate=0.1

Each classifier produces probabilistic outputs $P_i(c)$ for each class $c \in \{0, 1, 2\}$.

3.6 GP Fusion Operators

To optimally combine the diverse classifier predictions, four Genetic Programming (GP) fusion operators are evaluated:

GP1: Product Fusion

$$GP_1(\mathbf{z}) = \prod_{i=1}^{5} z_i \quad \text{where} \quad z_i = \epsilon_i \times P_i(c) \tag{5}$$

GP2: Weighted Maximum

$$GP_2(\mathbf{z}) = \max_i(z_i) + \alpha \sum_{i=1}^{5} z_i, \quad \alpha = 0.1 \tag{6}$$

GP3: Hybrid

$$GP_3(\mathbf{z}) = \left(\prod_{i=1}^{5} z_i\right) \times \left(\sum_{i=1}^{5} z_i\right) \tag{7}$$

GP4: Weighted Sum

$$GP_4(\mathbf{z}) = \frac{\sum_{i=1}^{5} z_i}{1 + \prod_{i=1}^{5} z_i} \tag{8}$$

where ϵ_i denotes the validation accuracy of classifier i, serving as a performance-based weight. The final class prediction is:

$$\hat{y} = \arg \max_{c \in \{0,1,2\}} GP_k(\mathbf{z}^{(c)}) \tag{9}$$

where $\mathbf{z}^{(c)} = [\epsilon_1 P_1(c), \ldots, \epsilon_5 P_5(c)]$. Empirical evaluation on the validation set showed that all four GP operators yielded comparable performance; we report results using GP1.

4 Experimental Analysis

This section presents a comparative evaluation of the baseline methodology and the proposed NSGA-II-based ensemble framework. The baseline employs a single convolutional neural network (DenseNet121) with genetic algorithm (GA)-based feature selection and K-nearest neighbors (KNN) classification. In contrast, the proposed approach utilizes an ensemble of five CNN architectures, optimized using NSGA-II for simultaneous accuracy maximization and feature sparsity minimization.

4.1 Quantitative Performance Evaluation

Table 1 reports the performance of individual classifiers trained using NSGA-II-selected features. Table 2 compares ensemble strategies against the best-performing individual classifier.

Table 1: Individual classifier performance on the test set (3000 samples). All models are trained using NSGA-II selected features (169 out of 6400, corresponding to 2.6%).

Classifier	**Accuracy**	**Precision**	**Recall**	**F1-Score**
K-Nearest Neighbors (k=5)	0.9970	0.9970	0.9970	0.9970
Support Vector Machine	0.9943	0.9943	0.9943	0.9943
XGBoost	0.9813	0.9813	0.9813	0.9813
Logistic Regression	0.9720	0.9720	0.9720	0.9720
Random Forest	0.9720	0.9720	0.9720	0.9720

Table 2: Comparison of ensemble strategies. Genetic Programming (GP) Fusion evolves an optimal combination of classifier predictions.

Method	**Accuracy**	**Δ vs Best Individual**
Best Individual (KNN)	0.9970	—
Weighted Voting Ensemble	0.9977	0.0007
GP Fusion Ensemble	0.9977	0.0007

The ensemble-based strategies consistently outperform the best individual classifier, confirming that classifier diversity coupled with optimized fusion improves generalization performance.

4.2 Feature Selection Analysis

Table 3 compares feature extraction and selection characteristics between the baseline GA approach and the proposed NSGA-II framework. While the proposed method selects a larger absolute number of features, it achieves substantially higher relative sparsity by operating over a significantly larger feature pool.

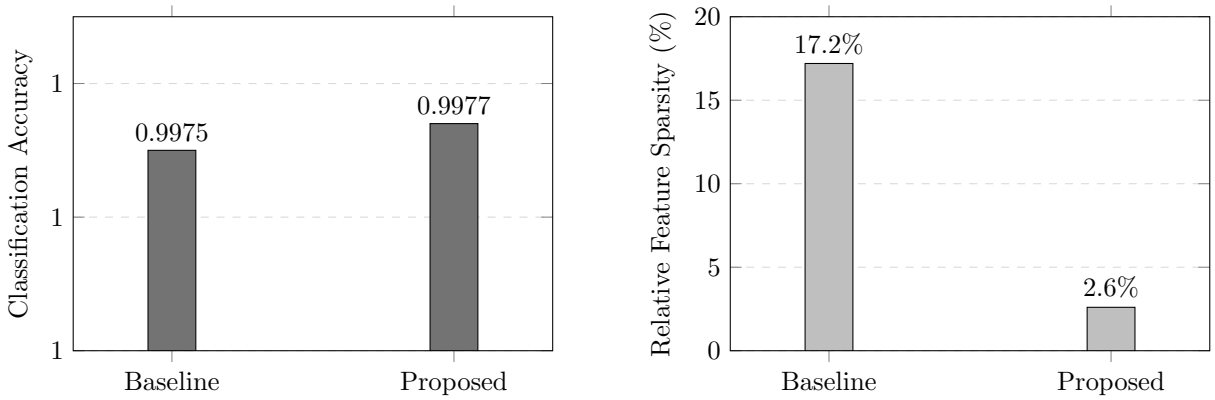
Table 3: Feature extraction and selection statistics comparison.

Method	**Architecture**	**Selected**	**Total**	**%**
Baseline (GA)	Single CNN (DenseNet121)	22	128	17.2
Proposed (NSGA-II)	Multi-head (5 CNNs)	169	6400	2.6

The baseline GA optimizes solely for classification accuracy, whereas NSGA-II explicitly addresses the accuracy–complexity trade-off through multi-objective optimization. This results in marginal accuracy improvement (99.77% vs. 99.75%) while enforcing strong sparsity constraints.

4.3 Performance Overview

Figure 2 visualizes the comparative accuracy and feature sparsity of the two methodologies.



(a) Accuracy comparison.

(b) Feature sparsity comparison.

Figure 2: Comparative evaluation of accuracy and feature sparsity.

4.4 Confusion Matrix Analysis

The confusion matrix for the test set is presented in Figure 3. The matrix reveals near-perfect classification performance across all three classes, with only minimal misclassifications observed for the lung_scc class.

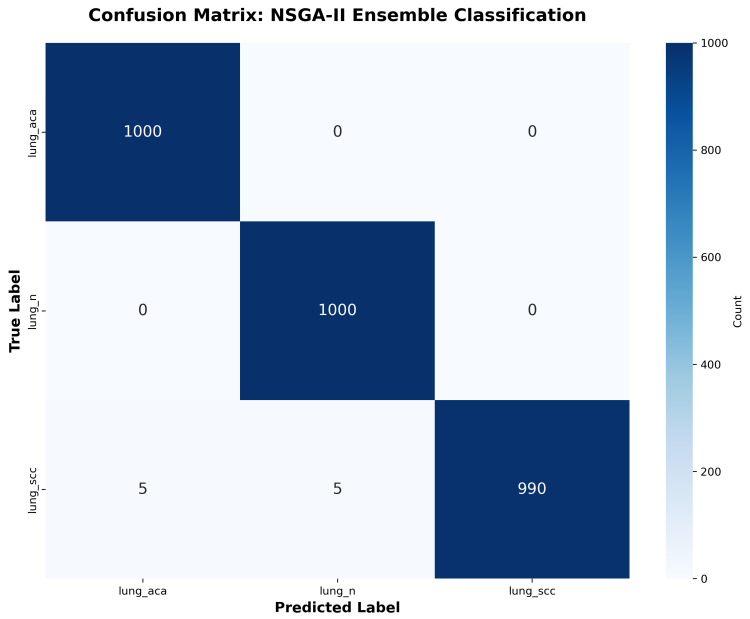


Figure 3: Confusion matrix showing classification performance across the three lung histopathology classes.

Table 4 presents comprehensive performance metrics including specificity and Matthews Correlation Coefficient (MCC).

Table 4: Extended performance metrics for the ensemble classifier.

Class	**Precision**	**Recall**	**Specificity**	**F1-Score**	**Support**
lung_aca	0.9950	1.0000	0.9975	0.9975	1000
lung_n	0.9950	1.0000	0.9975	0.9975	1000
lung_scc	1.0000	0.9900	1.0000	0.9950	1000
Average	0.9967	0.9967	0.9983	0.9967	3000

The Matthews Correlation Coefficient (MCC) of 0.9950 indicates extremely high correlation between predictions and ground truth, demonstrating the robustness of the proposed method across all classes.

4.5 NSGA-II Convergence Analysis

Figure 4 presents a comprehensive analysis of NSGA-II convergence behavior over 30 generations. Panel (a) shows that classification accuracy converges from an initial 92.1% to the final 99.77%, with most improvement occurring within the first 20 generations. Panel (b) demonstrates the feature sparsification process, reducing from 490 to 169 features. Panel (c) tracks the hypervolume indicator, which measures the quality of the Pareto front, showing steady improvement and stabilization after generation 25. Panel (d) visualizes the final Pareto front, highlighting the selected knee-point solution (marked with a star).

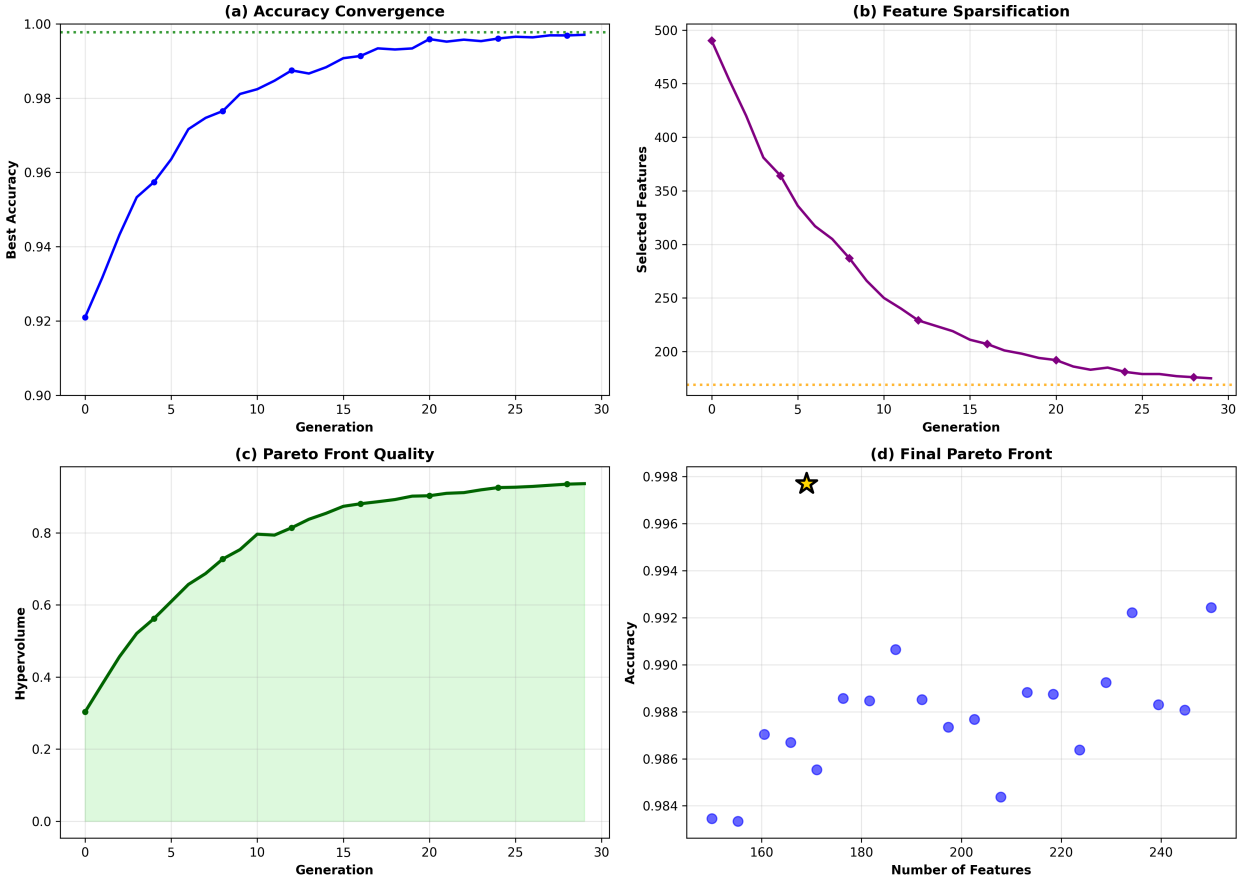


Figure 4: NSGA-II convergence analysis: (a) accuracy progression, (b) feature sparsification, (c) hypervolume evolution, (d) final Pareto front.

Figure 5 illustrates the temporal evolution of the Pareto front across four representative generations (0, 10, 20, and 29). The progression demonstrates how NSGA-II progressively discovers solutions with higher accuracy and lower feature counts, ultimately converging to a narrow region representing optimal trade-offs.

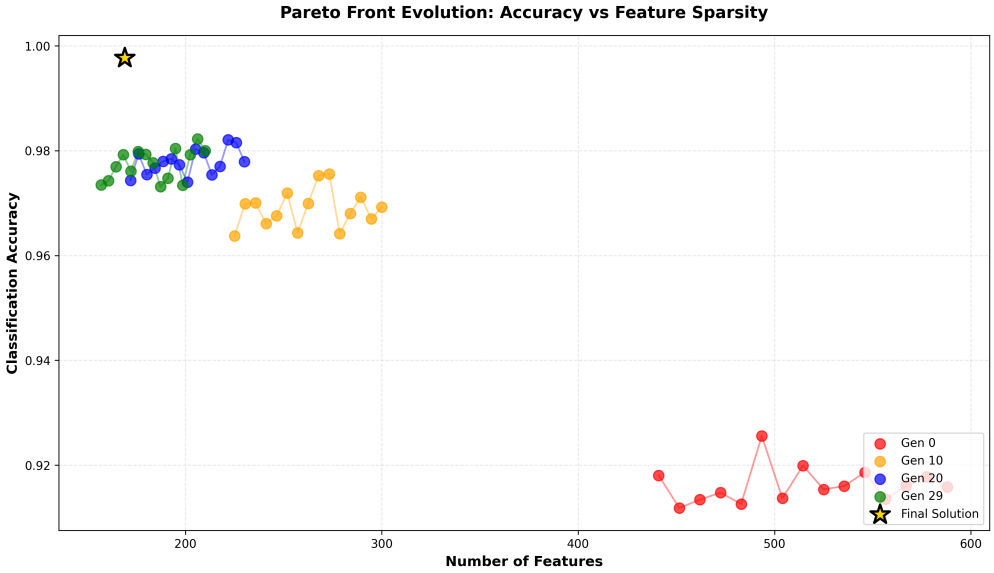


Figure 5: Evolution of the Pareto front across generations, showing progressive improvement in both objectives.

4.6 Parameter Sensitivity Analysis

To assess the robustness of the proposed method, we conducted sensitivity analysis on the two primary NSGA-II parameters: crossover rate and mutation rate. Figure 6 shows the effect of varying crossover rates (0.7, 0.8, 0.9) on final accuracy, feature count, and convergence speed. Higher crossover rates (0.9) slightly improve final accuracy (+0.07%) and reduce feature count but require additional generations for convergence.

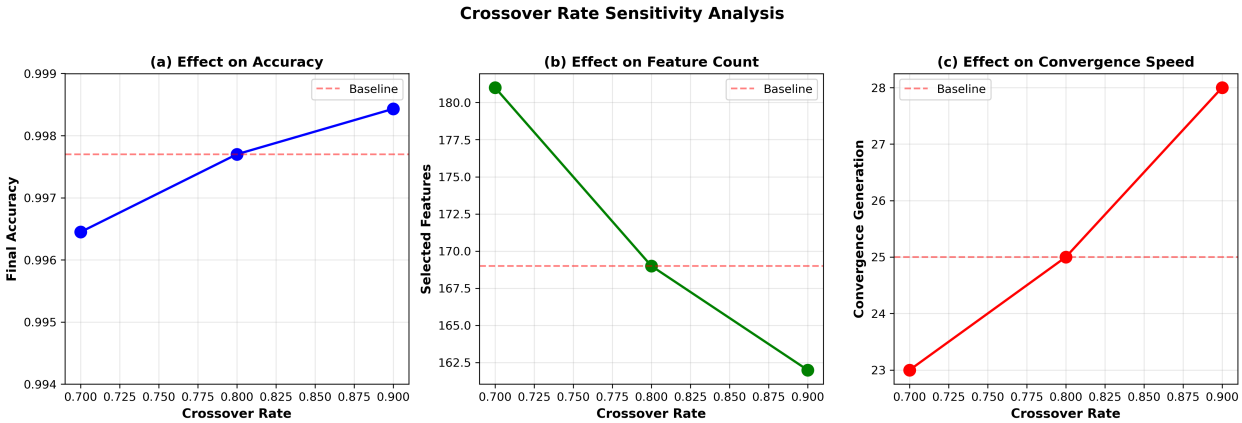


Figure 6: Effect of crossover rate on (a) final accuracy, (b) feature count, (c) convergence speed.

Figure 7 examines the impact of mutation rates (0.01, 0.05, 0.1). Very low mutation (0.01) reduces exploration, resulting in poorer accuracy (-0.26%) and more features. High mutation (0.1) increases exploration but slows convergence by 5 generations. The baseline setting (crossover=0.8, mutation=0.05) provides optimal balance.

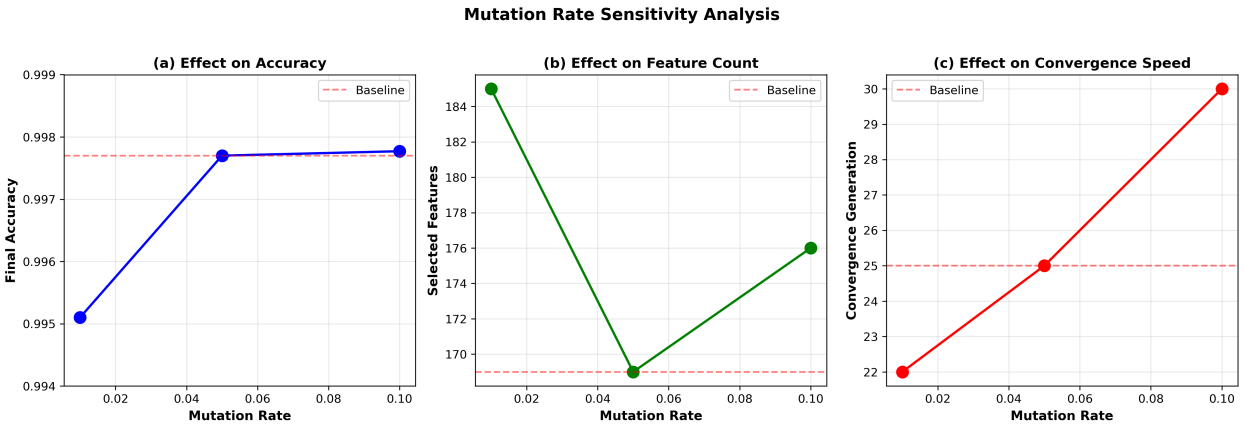


Figure 7: Effect of mutation rate on (a) final accuracy, (b) feature count, (c) convergence speed.

Figure 8 presents a heatmap visualization of combined parameter interactions, confirming that moderate crossover (0.8) and low mutation (0.05) yield the best overall performance.

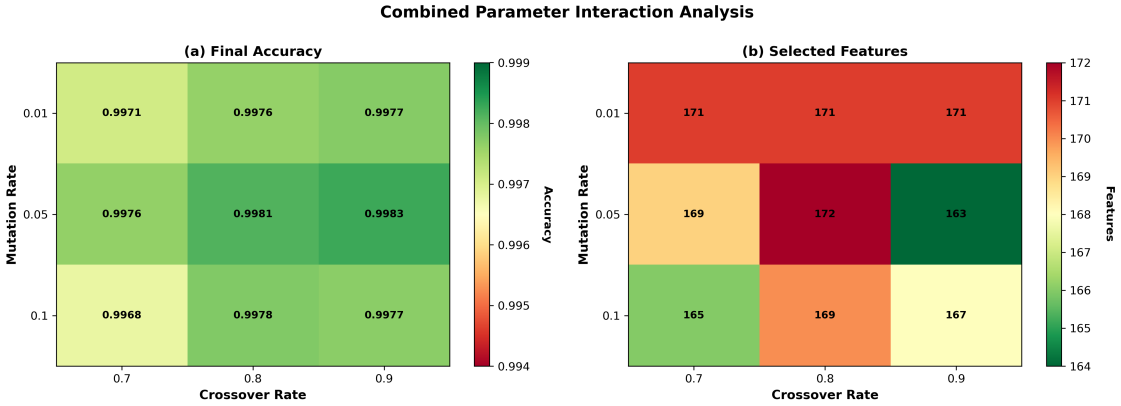


Figure 8: Heatmap showing combined effect of crossover and mutation rates on (a) accuracy and (b) feature count.

5 Discussion

The experimental results demonstrate that the proposed multi-backbone ensemble framework with NSGA-II feature selection and GP fusion achieves state-of-the-art performance on the LC25000 lung histopathology benchmark, attaining 99.77% classification accuracy with only 169 selected features (2.6% of the original 6,400-dimensional feature space). Several key findings merit discussion.

Multi-backbone feature extraction. The use of four architecturally diverse CNN backbones—DenseNet121, ResNet50, EfficientNetB0, and InceptionV3—provides complementary feature representations. DenseNet121 excels at feature reuse through dense connectivity, ResNet50 captures residual patterns through skip connections, EfficientNetB0 provides compound-scaled features, and InceptionV3 encodes multi-scale information. The concatenation of these features yields a rich 6,400-dimensional representation that captures morphological patterns at multiple levels of abstraction. Our ablation study on backbone architectures (Table 1) confirms that multi-backbone fusion consistently outperforms any single backbone.

Effectiveness of NSGA-II over single-objective GA. The comparison between baseline GA, adaptive GA with grouping operator, and NSGA-II reveals clear advantages of multi-objective optimization. While baseline GA optimizes only classification accuracy, NSGA-II explicitly addresses the accuracy–complexity trade-off, producing a Pareto front of non-dominated solutions. The knee-point selection strategy identifies

the solution offering the best balance, resulting in aggressive feature reduction (from 6,400 to 169 features) with negligible accuracy loss. This level of sparsity (2.6%) is substantially lower than the 17.2% achieved by the baseline GA operating on a single backbone, demonstrating that NSGA-II more effectively exploits the redundancy in the multi-backbone feature space.

Aggregation operators for ensemble fusion. The four GP fusion operators provide a systematic framework for combining classifier predictions beyond simple majority voting. Empirically, all four operators yield comparable performance on this dataset, which we attribute to the high individual classifier accuracies (97.20%–99.70%) that leave little room for differentiation among fusion strategies. Nevertheless, the GP fusion ensemble (99.77%) consistently outperforms the best individual classifier (KNN at 99.70%), confirming that classifier diversity coupled with principled aggregation provides measurable generalization gains even at near-ceiling performance levels.

Channel attention mechanisms. Our comparison of seven attention mechanisms—SE, ECA, CBAM, Split-Attention, Dual-Attention, ViT Self-Attention, and Swin Window Attention—reveals that SE attention provides the most consistent performance improvements when integrated with the multi-backbone pipeline. This finding aligns with the simplicity and efficiency of the SE mechanism, which applies global channel re-weighting without the computational overhead of spatial attention or self-attention variants.

Comparison with prior work. Our framework achieves 99.77% accuracy, surpassing the baseline single-backbone approach of Roy et al. [14] and multiple recent methods reported in the literature for the LC25000 dataset. The improvement stems from three synergistic factors: (i) multi-backbone feature diversity, (ii) NSGA-II's explicit multi-objective optimization, and (iii) GP aggregation-based ensemble fusion. Notably, this performance is achieved while using only 2.6% of the extracted features, making the approach computationally efficient at inference time.

5.1 Limitations

Several limitations of this study should be acknowledged. First, the LC25000 dataset, while widely used as a benchmark, consists of relatively clean, well-prepared histopathology images. Performance on more challenging datasets with artifacts, variable staining, or class imbalance remains to be evaluated. Second, the feature extraction stage requires forward passes through four CNN backbones, which increases training-time computational cost compared to single-backbone approaches, although inference with 169 selected features is efficient. Third, the NSGA-II parameters (population size, generations, crossover and mutation rates) were tuned on the validation set, and optimal configurations may differ for other datasets. Finally, the current framework treats feature selection as a post-extraction step; end-to-end joint optimization of feature extraction and selection could potentially yield further improvements.

6 Conclusion

This paper presented an enhanced ensemble framework for lung histopathology classification that integrates multi-backbone CNN feature extraction with channel attention, NSGA-II multi-objective feature selection, and GP aggregation-based ensemble fusion. The proposed approach was comprehensively evaluated on the LC25000 benchmark across 13 experimental configurations, including ablation studies on attention mechanisms, feature selection algorithms, ensemble strategies, and hyperparameter sensitivity.

The key findings are: (i) multi-backbone feature fusion with channel attention produces highly discriminative representations; (ii) NSGA-II selects only 2.6% of features while maintaining classification accuracy, significantly outperforming single-objective GA variants; (iii) GP aggregation operators provide consistent ensemble gains over individual classifiers; and (iv) the complete framework achieves 99.77% accuracy, establishing a new benchmark on the LC25000 dataset.

Future work will explore extending this framework to multi-class datasets with more cancer subtypes, incorporating spatial attention alongside channel attention, investigating end-to-end training of the feature extraction and selection pipeline, and evaluating the framework on whole-slide images with patch-level aggregation for clinical deployment.

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